

Selenium IDE Synchronization & Dynamic Waits Guide

Practical test script implementation for real-world scenarios

Task 1 Record Search & Wait for Flipkart Results

Test Execution Flow: Search for "Bluetooth Headphones" on Flipkart and pause until search results render before selecting the first product.

Command	Target	Value
open	/	
set window size	1280x800	
type	name=q	Bluetooth Headphones
sendKeys	name=q	\${KEY_ENTER}
wait for element visible	css=div[data-id]	10000
click	css=div[data-id] a	

Task 2 Add Explicit Wait for "Add to Cart" Button (Myntra)

Test Execution Flow: Ensure the "Add to Bag" element is fully rendered in the DOM and visible on-screen before attempting interaction.

Command	Target	Value
open	/path-to-product-page	
wait for element visible	xpath=//div[contains(text(),'ADD TO BAG')]	5000
click	xpath=//div[contains(text(),'ADD TO BAG')]	

Task 3 Handle AJAX Dynamic Dropdown (Zomato)

Test Execution Flow: Type a location into the search box and wait for the dynamic AJAX suggestion container to load before clicking the top option.

Command	Target	Value
open	/	
type	css=input[placeholder*='Search']	Mumbai
wait for element visible	css=div[class*='search-dropdown']	8000
click	css=div[class*='search-dropdown'] p:first-child	

Task 4 Fix Flaky Instagram Login Button Delay

Root Cause Analysis: Instagram uses client-side rendering (React/Next.js). The DOM inserts elements instantly, but JavaScript hydration causes a delay before the submit button becomes interactive. Clicking too quickly leads to intermittent test failures.

Reliable Test Script:

Command	Target	Value
open	/accounts/login/	
type	name=username	my_user
type	name=password	my_pass
wait for element visible	css=button[type='submit']	5000
wait for element editable	css=button[type='submit']	5000
click	css=button[type='submit']	

Task 5 Implicit vs. Explicit Waits in Selenium

Key Differences:

- **Implicit Wait:** A global configuration applied to the entire driver instance. It tells WebDriver to continuously poll the DOM for a specified timeframe before throwing a `NoSuchElementException` for *any* missing element.
- **Explicit Wait:** A targeted wait applied to a *specific element* under custom conditions (e.g., waiting for visibility, editability, or specific text) before proceeding.

Real-World Applications:

- **Implicit Wait Example (WhatsApp Web):** Applied globally when opening the application so WebDriver gives static elements (like the search bar or chat list) up to 10 seconds to load across the entire session.
- **Explicit Wait Example (YouTube Uploads):** The "Publish" button is already rendered in the DOM but stays disabled while a video processes. An explicit wait (`wait for element editable`) specifically targets the "Publish" button until processing finishes.